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**A Data Science Workflow for Strengthening Tribal Public Health Surveillance:
Development and Evaluation of a Social Determinants of Health Dashboard and Training
Framework**

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Abstract

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Indigenous public health systems operate within structurally constrained data environments characterized by small population sizes, frequent data suppression, racial misclassification, and fragmentation across Tribal, state, and federal data sources.

Structural data challenges, current approaches to training and education within Tribal public health systems remain limited in their ability to support methodological understanding of these issues. These limitations are further compounded by misalignment between Tribal and state public health systems, where data are often produced, defined, and disseminated within state or federal frameworks that do not reflect Tribal governance. Training is often episodic, externally delivered, and focused on the use of specific tools or platforms rather than on developing a conceptual understanding of data limitations such as small-area instability, measurement error, and cross-source inconsistency. Consequently, there is a disconnect between tool use and methodological understanding. As a result, Tribal public health analysts may be expected to use and interpret complex and imperfect data without the training needed to diagnose bias, assess uncertainty, or evaluate the implications of integrating multiple data sources. Critically, there are few accessible frameworks or tools that enable Tribal public health departments to not only recognize these limitations, but to engage with them in a meaningful analytic way.

In response, this study develops and evaluates an applied data science workflow for strengthening Tribal public health surveillance under real-world data constraints. In collaboration with Cherokee Nation Public Health (CNPH), we developed an interactive R Shiny dashboard that harmonizes county-level Social Determinants of Health (SDOH) data from federal (ACS5, USDA), state (OK2Share), and Tribal (CNPH) sources, organizing 20 indicators into six SDOH domains: Economic Instability, Education Attainment, Housing and Transportation Barriers, Health Care Access, Environmental Burden, and Social and Community Context. A structured training and capacity-building framework was embedded directly within the dashboard and evaluated through a mixed-methods with five CNPH analysts. A strong overall usability (mean rating 4.33 out of 5.0) and high ratings for map navigation and SDOH domain clarity. A critical finding was the distinction between usability and interpretability: Interpretation of standardized metrics—including z-scores, composite domain scores, and percentile rankings—presented greater difficulty (mean score 3.67), underscoring that accessible tools do not automatically produce analytic understanding. The resulting framework provides a practical and theoretically grounded model for locally governed, data-driven public health decision-making in Tribal settings, demonstrating not only data integration and visualization, but also embedded training, interpretability support, and iterative refinement aligned with Indigenous Data Sovereignty principles.

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1. Introduction

Indigenous American Indian and Alaska Native (AI/AN) Tribal populations experience persistent inequities in public health data availability, quality, and use, limiting the ability to generate accurate and actionable information to support community health decision-making (Field et al., 2024; Carroll et al., 2020; Rainie et al., 2019). These inequities are not solely the result of data limitations, but reflect deeper structural challenges in how data systems are designed, governed, and operationalized within and across Tribal, state, and federal contexts.

In the United States, multiple factors contribute to these data inequities. Small population sizes often lead to statistical instability and frequent data suppression in publicly available datasets (Rao and Molina, 2015; Datta and Ghosh, 2012; Wakefield, 2020). At the same time, racial misclassification in administrative and vital records has been consistently documented, resulting in systematic underestimation of morbidity and mortality burdens among AI/AN populations (Espey et al., 2014; Arias et al., 2016; Norris et al., 2012; Jim et al., 2014; Rodriguez-Lonebear, 2021). Data relevant to Tribal populations are further fragmented across federal surveys, state administrative systems, and Tribal health records, each with distinct definitions, coverage, and accessibility (Dahdouh et al., 2019; Jiang et al., 2021; Little and Rubin, 2002). Together, these factors produce sparse, incomplete, and often inconsistent population health estimates, particularly at local and small-area levels.

Existing surveillance infrastructures are typically organized around federal and state reporting requirements, rather than the jurisdictional realities and governance structures of sovereign Tribal Nations (Kitchin, 2022; Benitez et al., 2020). As a result, data systems often fail to align with Tribal governance, priorities, and decision-making needs, limiting their utility within Tribal public health contexts. Similar patterns are observed globally across Indigenous populations, where misalignment between data systems and Indigenous governance structures contributes to persistent inequities in data availability and use (Walter and Suina, 2019; Lovett et al., 2016; Gartner et al., 2023).

In response, there has been increasing recognition of Indigenous Data Sovereignty (IDS),

which asserts the rights of Indigenous Peoples to control the collection, access, ownership, and application of data concerning their communities (Kukutai and Taylor, 2016; Carroll et al., 2020; Rhodes et al., 2024). IDS emphasizes that data are not neutral, but are embedded within systems of power, governance, and control (Walter and Suina, 2019; Smith, 2016). The CARE principles — Collective benefit, Authority to control, Responsibility, and Ethics — provide a framework for operationalizing IDS in practice, emphasizing the need for data systems that support Tribal governance, build local analytic capacity, and enable responsible data use aligned with community priorities (Carroll et al., 2020, 2021).

Despite these advances, three interrelated challenges remain largely unaddressed. First, data fragmentation, statistical instability, and governance misalignment continue to constrain the production of reliable, locally relevant population health estimates in Tribal settings (Wakefield, 2020; Rao and Molina, 2015; Jiang et al., 2021). Second, while a substantial statistical literature has developed methods to address data sparsity, bias, and multi-source integration, these approaches are rarely translated into practice due to data access constraints, computational requirements, and limited local analytic infrastructure (Wakefield, 2020; Mercer et al., 2016; Banerjee et al., 2014; Gelman et al., 2013). Third, training within Tribal public health systems remains episodic, externally delivered, and focused on specific tools rather than on developing a conceptual understanding of data limitations — creating a persistent disconnect between tool use and methodological understanding (Duarte, 2021; Rainie et al., 2019; Carroll et al., 2021; Brownson et al., 2018; Frenk et al., 2010). As a result, Tribal public health analysts may be expected to use and interpret complex and imperfect data without the training needed to diagnose bias, assess uncertainty, or evaluate the implications of integrating multiple data sources.

Existing approaches tend to address these challenges in isolation — advancing methodological tools without attention to implementation, or building data systems without integrating the training needed to sustain them. There is comparatively little work that integrates data harmonization, visualization, and capacity building within a unified framework aligned with Tribal governance structures and practical data constraints (Duarte, 2021; Fixsen et al., 2005; Nilsen, 2015). This

study addresses that gap by developing an applied data science workflow that integrates multi-source SDOH data, interactive visualization, and structured training within a Tribal public health system, developed in collaboration with Cherokee Nation Public Health (CNPH). The dashboard component addresses the first two challenges by harmonizing county-level data from federal, state, and Tribal sources within a descriptive, operationally feasible framework; the embedded training component directly addresses the third by integrating capacity building into routine data use rather than treating it as a separate activity. The approach emphasizes usability, transparency, and alignment with IDS principles.

Critical to this development is the direct input of Tribal public health analysts, data stewards, and community stakeholders, whose lived experience and contextual knowledge are essential for ensuring that data systems are interpretable, relevant, and aligned with Tribal governance and priorities (Carroll et al., 2020; Rainie et al., 2019; Rhodes et al., 2024). Although developed within the Cherokee Nation context, the proposed framework is designed to be adaptable to other Tribal Nations and Indigenous populations facing similar challenges of data fragmentation, governance misalignment, and limited analytic infrastructure (Walter and Suina, 2019; Field et al., 2024).

The specific objectives of this study were: (1) to develop a county-level SDOH surveillance dashboard for Cherokee Nation Public Health that harmonizes multi-source data from federal, state, and Tribal systems into an accessible, interactive analytic platform; (2) to embed a structured training and capacity-building framework directly within the dashboard workflow; and (3) to evaluate the usability, interpretability, and practical utility of the integrated system through a mixed-methods assessment with CNPH analysts.

Figure 1 summarizes the structure of the thesis, illustrating the progression from conceptual framing of Indigenous data systems (Section 2) to the development of an integrated analytic and capacity-building workflow (Section 3), followed by evaluation through applied use and user feedback (Section 4), and interpretation of findings and broader implications for Tribal public health systems (Section 5).

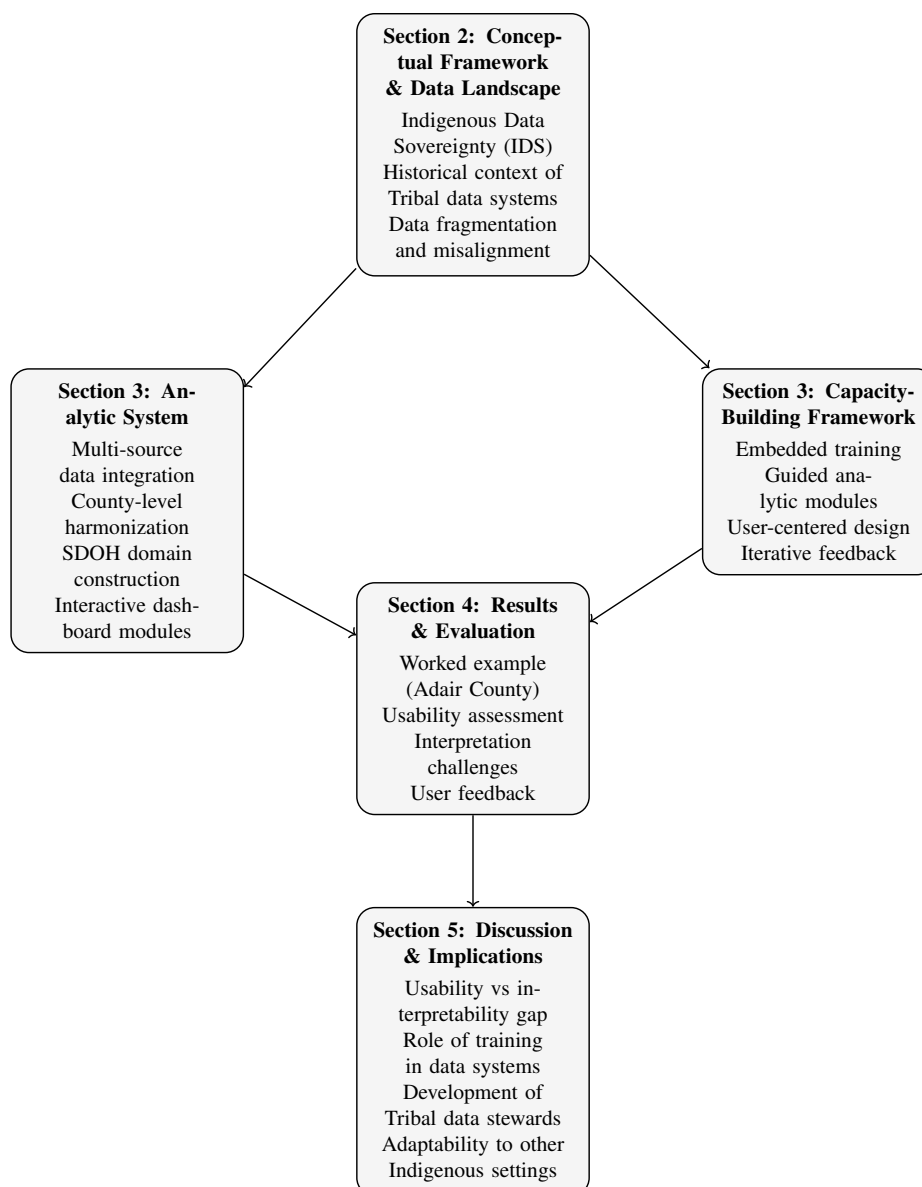


Figure 1: Thesis roadmap illustrating the progression from conceptual framing of Indigenous data systems (Section 2) to the development of an applied analytic and capacity-building framework (Section 3), followed by evaluation of system usability and interpretability (Section 4), and interpretation of findings and broader implications for Tribal public health systems (Section 5).

2. Conceptual Framework and Data Landscape in Tribal Health Systems

2.1. *Historical Context of Tribal Data Systems and the Emergence of Indigenous Data Sovereignty*

Understanding contemporary data inequities in Tribal health surveillance requires situating them within the historical development of data systems affecting Indigenous populations. Prior to the emergence of Indigenous Data Sovereignty (IDS), data concerning Indigenous communities were largely collected, controlled, and interpreted by external entities, including federal governments, state agencies, and academic institutions (Smith, 2016; Walter and Suina, 2019; Field et al., 2024). These systems were not designed to reflect Tribal governance structures or community priorities, but instead to serve administrative and regulatory functions rooted in colonial governance frameworks (Kukutai and Taylor, 2016; Lovett et al., 2016; O'Donnell et al., 2016).

Historically, Indigenous populations were inconsistently enumerated or excluded from early census and vital registration systems, contributing to long-standing gaps in demographic and health data (Thornton, 1997; Snipp, 2016; Anderson et al., 2016). Even as national statistical systems expanded, Indigenous identity was frequently misclassified or aggregated into residual categories, obscuring heterogeneity across Tribal Nations and limiting visibility in official statistics (Snipp, 2016; Walter and Suina, 2019; Smith, 2016). These patterns reflect broader forms of statistical marginalization, in which data systems reproduce existing power structures and inequalities (Anderson et al., 2016; Singh et al., 2024).

In response to these limitations, multiple health data and service systems have emerged that differ in governance, scope, and alignment with Tribal priorities. The Indian Health Service (IHS) is a federal agency responsible for providing healthcare services to American Indian and Alaska Native populations, operating facilities and maintaining administrative data systems within a federal framework (Sequist, 2011; Reddy, 2016). Tribal Epidemiology Centers (TECs), while often Tribally affiliated and regionally based, are federally funded public health authorities that support surveillance, epidemiologic analysis, and technical assistance across Tribal regions, and function within broader federal public health infrastructure (Goodman, 2012; Rainie et al., 2019). In contrast, Tribal health departments operated under the Indian Self-Determination and Education

Assistance Act (Public Law 93-638) represent locally governed systems in which Tribal Nations assume authority over the administration of federally funded health programs (U.S. Congress, 1975; Reddy, 2016; Sequist, 2011). These “638” systems, such as Cherokee Nation Public Health, are designed and managed by Tribal governments and are more directly aligned with community priorities, service delivery contexts, and Tribal governance structures.

Despite these advances, no single system provides comprehensive coverage of Tribal populations. IHS and TEC data systems operate within federal frameworks and may not fully capture local context or populations receiving care outside those systems, while Tribal health departments—although locally governed—may face limitations in coverage, infrastructure, and interoperability with external data sources (Rainie et al., 2019; Carroll et al., 2021; AbouZahr et al., 2019; World Health Organization, 2018). This multi-system structure reflects both progress toward Tribal self-determination and the continued fragmentation of data across federal and Tribal domains.

This has led to the emergence of a dual-system structure in Indigenous health data. Tribal data systems are governed by sovereign Tribal Nations and prioritize community relevance and data stewardship, while state and federal systems are designed for large-scale surveillance and policy development under broader regulatory frameworks (Rainie et al., 2019; Field et al., 2024; Kitchin, 2022). Neither system alone provides complete or accurate coverage of Tribal populations.

The emergence of IDS reflects a critical response to these historical and structural challenges. IDS asserts that data concerning Indigenous communities are not neutral, but are embedded within systems of power, governance, and control (Kukutai and Taylor, 2016; Rhodes et al., 2024). The CARE principles—Collective benefit, Authority to control, Responsibility, and Ethics—provide a framework for operationalizing IDS in practice (Carroll et al., 2020, 2021). Within public health, IDS emphasizes the need for accurate, robust, and comprehensive data systems that support Tribal governance, build local analytic capacity, and enable responsible data integration that respects sovereignty and community priorities (Rainie et al., 2019; Duarte, 2021). This context motivates the need for applied data science frameworks that not only integrate fragmented data sources, but also embed interpretability, training, and governance-aligned analytic practices within Tribal

public health systems.

2.2. Data Limitations and Inequities in Tribal Health Surveillance

In practice, these challenges are most evident within public health surveillance systems, where issues of governance, representation, and data control directly shape the production of population health statistics. Despite their central role in disease monitoring, resource allocation, and policy development, a substantial body of literature demonstrates that these systems systematically underrepresent and misreport Indigenous populations in both the United States and globally (Espey et al., 2014; Arias et al., 2016; Gartner et al., 2023).

Tribal data systems typically capture individuals who access Tribal or IHS services and may not include those receiving care outside these systems, resulting in incomplete population coverage (Sequist, 2011; Reddy, 2016; Gracey and King, 2009). At the same time, state and federal data systems provide broader geographic coverage but often lack accuracy and granularity for AI/AN populations due to misclassification, aggregation, and inconsistent reporting practices (Espey et al., 2014; Gartner et al., 2023; ?). Together, these coverage limitations mean that no single data source provides a complete or unbiased picture of AI/AN population health.

Among AI/AN populations, racial misclassification in administrative and vital records is pervasive and represents a primary source of bias in health and mortality estimates (Espey et al., 2014; Arias et al., 2016; Norris et al., 2012; Jim et al., 2014). Linkage studies between national datasets and IHS records have shown that misclassification occurs at rates approaching 30% leading to systematic underestimation of morbidity and mortality burdens; correcting for this misclassification can substantially alter estimates of AI/AN health disparities (Espey et al., 2014; Jim et al., 2014).

These measurement errors are compounded by statistical challenges associated with small population sizes. AI/AN populations often constitute a small proportion of the total population within geographic units, resulting in unstable estimates, high variance, and frequent data suppression in publicly released datasets (Rao and Molina, 2015; Datta and Ghosh, 2012; Mercer et al., 2016; Wakefield, 2020). While suppression practices are intended to protect confidentiality, they substantially reduce data availability and obscure meaningful spatial and temporal patterns in AI/AN

health outcomes (Gomez et al., 2014; Benitez et al., 2020; Jiang et al., 2021). As a result, many Tribal areas lack publicly reported data on key health indicators. In addition to instability and suppression, aggregation practices further contribute to inequities in representation. Indigenous populations are frequently grouped into broad racial or ethnic categories, masking heterogeneity across Tribal Nations and limiting the identification of localized disparities (Gartner et al., 2023; Lovett et al., 2016; O'Donnell et al., 2016). These practices reflect structural features of surveillance systems that prioritize national comparability over local relevance and community-specific insights.

These methodological challenges are not only technical, but interpretive. In practice, Tribal public health analysts must make sense of data that are simultaneously sparse, suppressed, misclassified, and distributed across multiple systems with differing definitions and coverage. Interpreting such data requires more than descriptive analysis; it requires the ability to recognize when observed patterns reflect underlying population processes versus artifacts of measurement error, suppression rules, or data integration decisions (Chen et al., 2011; Jiang et al., 2021). For example, distinguishing true geographic variation from instability due to small population sizes, or understanding how linkage-based corrections for racial misclassification alter estimates, demands familiarity with the assumptions and limitations of the underlying methods (Wakefield, 2020; Rao and Molina, 2015; Norris et al., 2012; Jim et al., 2014). Without this methodological context, analytic outputs may be treated as fixed representations of population health rather than as estimates shaped by data limitations and modeling choices (Gelman et al., 2013; Bhattacharya et al., 2019).

2.3. Data Availability and Governance Constraints in Tribal Health Systems

Public health data relevant to AI/AN populations are distributed across multiple systems, including federal surveys, state administrative datasets, and Tribal health records, each characterized by differing definitions, coverage, and accessibility. Information relevant to AI/AN populations is distributed across these systems, each with distinct definitions, coverage, and accessibility (Little and Rubin, 2002; Kitchin, 2022; Dahdouh et al., 2019). This lack of interoperability constrains the construction of coherent analytic datasets and limits the ability to conduct longitudinal and small-

area analyses (Chen et al., 2011; Jiang et al., 2021). Barriers to data sharing across Tribal, state, and federal systems—including legal, ethical, and technical constraints—further limit integration, and methodological approaches for combining these data while respecting governance constraints remain underdeveloped (Jiang et al., 2021; Dahdouh et al., 2019; Duarte, 2021; ?; Kitchin, 2022; Benitez et al., 2020).

In practice, access to these data is not uniform. Publicly available datasets, such as federal surveys, provide broad geographic coverage but often lack specificity for AI/AN populations, while Tribal and IHS data systems may offer more relevant and accurate information but are subject to governance, approval processes, and access restrictions that limit their availability for analysis (Carroll et al., 2021). As a result, analytic work in Tribal public health contexts is shaped not only by what data exist, but by what data can be accessed, approved, and used within specific governance frameworks.

Within this context, data integration is necessarily an applied and pragmatic process. Rather than relying on fully linked datasets, analysts must work with partially overlapping data sources that differ in resolution, population coverage, and accessibility. This requires approaches that emphasize harmonization across available data, careful consideration of geographic definitions, and explicit recognition of the limitations introduced by incomplete coverage and restricted access. These constraints directly shape both the selection of indicators and the structure of analytic frameworks. Indicators must be constructed from data that are consistently available and approved for use, often requiring the combination of AI/AN-specific measures with broader population indicators. Similarly, geographic aggregation is frequently determined by the level at which data can be consistently accessed across sources, rather than by an ideal level of spatial resolution.

These conditions underscore that data integration in Tribal public health settings is not solely a technical problem, but a governance-driven and access-constrained process. As a result, analytic frameworks must balance methodological rigor with feasibility, while remaining aligned with Tribal data governance and the practical realities of working across partially accessible systems. This context motivates the need for applied analytic approaches that explicitly incorporate data

availability, access constraints, and interpretability into their design, rather than treating these factors as secondary considerations.

2.4. Workflow Development and Capacity Building in Applied Tribal Public Health Data Systems

The development of applied data systems in public health is an inherently implementation-dependent process. Evidence from implementation science indicates that the adoption and sustained use of analytic tools is shaped not only by their technical design, but by the organizational contexts, workforce capabilities, and training structures within which they are deployed (Fixsen et al., 2005; Nilsen, 2015; Brownson et al., 2018). Systems developed without attention to local context, usability, and capacity are frequently underutilized or abandoned following initial deployment, regardless of their technical quality (Braa et al., 2004; Brownson et al., 2018). This implementation gap — the distance between the availability of a tool and its meaningful use in practice — represents a persistent challenge in applied public health data systems (Nilsen, 2015; Frenk et al., 2010).

A central contributor to this gap is the mismatch between the analytic complexity of modern public health data systems and the training available to the analysts expected to use them. Public health surveillance increasingly relies on multi-source data integration, spatial analysis, and model-based estimation, yet formal training in these methods remains limited within many local and Tribal public health departments (Brownson et al., 2018; Frenk et al., 2010; Drehoobl et al., 2012). As a result, analysts may be expected to interpret and act on complex outputs without the methodological background needed to assess their reliability, recognize artifacts of data limitation, or adapt analytic approaches to local contexts (Bhattacharya et al., 2019; Chen et al., 2011). This disconnect between tool use and methodological understanding reduces the practical impact of analytic systems and limits their contribution to evidence-informed decision-making.

Within Tribal public health settings, these challenges are further amplified by structural conditions including limited staffing, constrained infrastructure, and historical reliance on analytic products developed and maintained by external academic or governmental partners (Duarte, 2021; Rainie et al., 2019; Carroll et al., 2021). Externally developed systems may not align with local

governance structures, data definitions, or community priorities, and their long-term sustainability is often contingent on continued external involvement rather than internal capacity (Braa et al., 2004; Wallerstein and Duran, 2010). Developing locally grounded analytic capacity — the ability to manage, interpret, adapt, and sustain data systems from within Tribal institutions — is therefore not only a technical priority but a governance imperative, directly supporting Indigenous Data Sovereignty by enabling Tribal Nations to exercise meaningful control over how data are operationalized and applied (Carroll et al., 2020; Rainie et al., 2019; Rhodes et al., 2024).

Addressing these challenges requires integrating capacity building directly within the analytic workflow, rather than treating training as a supplementary activity delivered separately from system use. Approaches grounded in adult learning theory emphasize experiential, problem-centered instruction anchored in immediate real-world application as more effective than episodic or platform-specific training (Knowles et al., 2015; Kolb, 1984; Merriam, 2007). Participatory and community-engaged frameworks similarly support the co-development of analytic tools and training with the communities they are intended to serve, ensuring that both the system and the interpretive skills needed to use it are locally grounded and sustainable (Wallerstein and Duran, 2010; Duarte, 2021). Embedding training within the data workflow supports iterative learning, contextual interpretation, and sustained local ownership over time (Fixsen et al., 2005; Nilsen, 2015).

This study addresses this gap by developing an integrated data science workflow that embeds capacity building, interpretability, and training directly within an applied SDOH surveillance system, developed in collaboration with Cherokee Nation Public Health and aligned with the governance and analytic priorities of Tribal public health practice.

3. Methods

3.1. Overview

We developed a data science workflow for small-area SDOH surveillance in Tribal health systems, implemented within the Cherokee Nation Public Health (CNPH) context. The workflow consists of three integrated components: (i) computational abilities, encompassing multi-source data integration, statistical methods, and the R Shiny platform; (ii) educational training, including a structured curriculum, focus group facilitation, and capacity-building grounded in Indigenous Data Sovereignty principles; and (iii) translational tools, comprising the interactive dashboard modules that bridge data analysis and public health decision-making. The workflow is grounded in principles of IDS, emphasizing local control, contextual interpretation, and appropriate use of data within Tribal governance systems (Carroll et al., 2020; Kukutai and Taylor, 2016). Figure 3.2.4 illustrates the three-component architecture of this workflow.

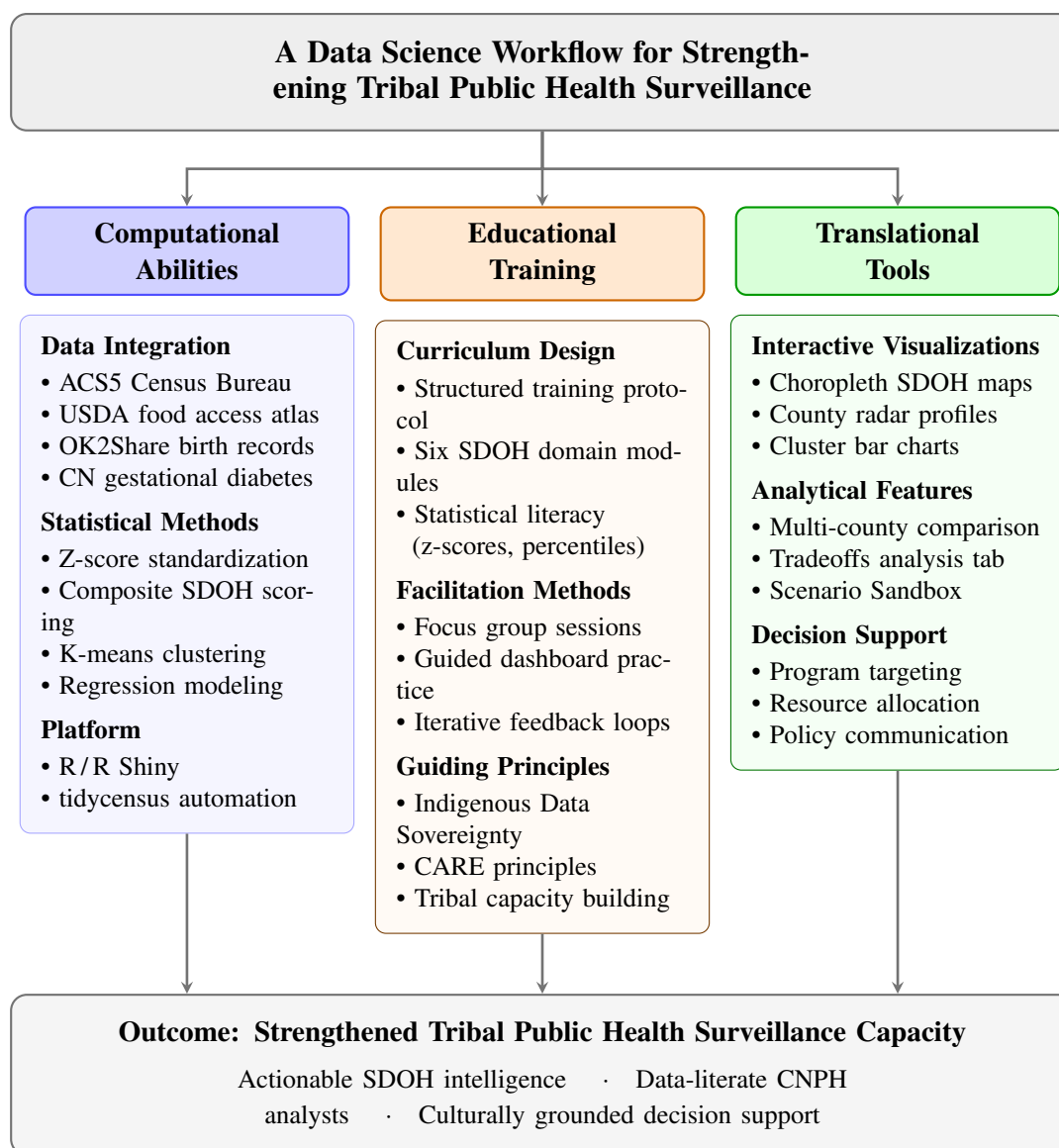


Figure 2: Three-component methodological framework for the Cherokee Nation SDOH data science workflow. Computational abilities (data integration, statistical methods, and R/Shiny platform), educational training (curriculum design, facilitation, and Indigenous Data Sovereignty principles), and translational tools (interactive visualizations, analytical features, and decision support) operate in concert to produce strengthened Tribal public health surveillance capacity.

3.2. Development of an Interactive SDOH Dashboard for Tribal Health Systems

3.2.1. Data Sources and Integration

The analytic system integrates data from multiple sources spanning federal, state, and Tribal systems. These include the American Community Survey (ACS) 5-year estimates, a nationally representative survey conducted by the U.S. Census Bureau that provides detailed demographic,

socioeconomic, and housing characteristics at small geographic scales (U.S. Census Bureau, 2022). State-level data are obtained from OK2SHARE, a public health data system maintained by the Oklahoma State Department of Health that aggregates vital statistics, hospital discharge data, and behavioral health indicators for surveillance and reporting purposes (Oklahoma State Department of Health, 2022). In addition, administrative surveillance data maintained by Cherokee Nation Public Health (CNPH) are incorporated, which include programmatic and clinical data specific to Tribal populations and reflect healthcare utilization and outcomes within Tribal service systems (Cherokee Nation Public Health, 2022). These sources differ in sampling design, population coverage, and reporting structure. As a result, they differ in accuracy, coverage, and level of aggregation. ACS data are derived from a rolling sample survey, while state administrative systems aggregate registry and reporting data. Tribal administrative data are population-specific but may not capture individuals receiving care outside Tribal systems. As a result, the integrated dataset reflects a setting in which no single data source provides complete population coverage (Little and Rubin, 2002). To enable integrated analysis, all data were harmonized to a common spatial unit (county level) and aligned temporally to a common reference year (2022). Geographic identifiers were standardized using Federal Information Processing Standards (FIPS) codes, and variables were recoded to ensure conceptual consistency across data sources.

3.2.2. SDOH Domain Construction

To support a structured and multidimensional assessment of social determinants of health, county-level indicators were organized into six domains representing key structural and environmental factors influencing population health. These domains include Economic Instability, Education Attainment, Housing and Transportation Barriers, Health Care Access, Environmental Burden, and Social and Community Context. Each domain captures a distinct dimension of the broader social and environmental conditions that shape health outcomes.

Indicators were selected based on two primary criteria: (i) availability across multiple data sources with consistent county-level coverage, and (ii) relevance to Tribal public health priorities. Where available, both general population indicators (e.g., poverty, unemployment, housing con-

ditions) and AI/AN-specific measures (e.g., education distribution and insurance coverage among AI/AN populations) were included. This dual inclusion enables simultaneous assessment of overall population conditions and disparities affecting Tribal populations.

All indicators were harmonized to a common spatial unit (county level) to enable consistent geographic comparison across Oklahoma. Differences in measurement across data sources were addressed through variable recoding and alignment to ensure conceptual comparability. Table 1 summarizes the grouping of indicators within each domain, while Table 2 provides detailed definitions and data sources for each variable.

Table 1: Mapping of SDOH Domains to Component Indicators

Domain	Component Indicators
Economic Instability	Poverty rate; unemployment rate; SNAP participation; low-income households
Education Attainment	Less than high school; high school graduate; bachelor's degree or higher; AI/AN education distribution
Housing and Transportation Barriers	Cost-burdened renters; overcrowded housing; no vehicle access; incomplete plumbing
Health Care Access	Uninsured rate; Medicaid coverage; AI/AN insurance coverage; access to Indian Health Service (IHS)
Environmental Burden	PM2.5 exposure; low access to parks or open space
Social and Community Context	Disability rate; single-parent households; racial composition; AI/AN population proportion

The indicators defined above serve as the basis for domain-level SDOH measures. By grouping related variables into conceptually coherent domains, this approach reduces dimensionality while preserving meaningful variation across social and environmental conditions. The inclusion of both general population and AI/AN-specific indicators further supports interpretation of disparities within the broader population context.

3.2.3. Standardization and Composite Score Construction

To facilitate comparability across counties and indicators, all variables were standardized using z-scores computed across Oklahoma counties:

$$z = \frac{x - \mu}{\sigma}$$

Table 2: Social Determinants of Health (SDOH) Indicators, Definitions, and Data Sources

Domain	Indicator	Definition	Source
Economic Instability	Poverty Rate	Percentage of individuals living below the federal poverty level	ACS
	Unemployment Rate	Percentage of the civilian labor force that is unemployed	ACS
	SNAP Participation	Percentage of households receiving Supplemental Nutrition Assistance Program benefits	ACS
Education Attainment	Less than High School	Percentage of adults aged 25+ without a high school diploma	ACS
	High School Graduate	Percentage of adults aged 25+ with a high school diploma or equivalent	ACS
	Bachelor's Degree or Higher	Percentage of adults aged 25+ with a bachelor's degree or higher	ACS
	AI/AN Education Distribution	Distribution of educational attainment among AI/AN populations	ACS
Housing and Transportation	Cost-Burdened Renters	Percentage of renter households spending more than 30% of income on housing	ACS
	Overcrowded Housing	Percentage of households with more than one person per room	ACS
	No Vehicle Access	Percentage of households without access to a vehicle	ACS
	Inadequate Plumbing	Percentage of households lacking complete plumbing facilities	ACS
Health Care Access	Uninsured Rate	Percentage of individuals without health insurance coverage	ACS
	Medicaid Coverage	Percentage of individuals enrolled in Medicaid	ACS
	IHS Utilization Proxy	Percentage of AI/AN individuals relying on Indian Health Service coverage	ACS / CNPH
Environmental Burden	PM2.5 Exposure	Annual average concentration of fine particulate matter (PM2.5)	EPA / ACS-linked
	Low Access to Parks	Percentage of population with limited access to parks or recreational space	ACS / external
Social and Community Context	Disability Rate	Percentage of individuals with a reported disability	ACS
	Single-Parent Households	Percentage of households headed by a single parent	ACS
	Racial Composition	Distribution of population by race/ethnicity categories	ACS
	AI/AN Population Proportion	Percentage of population identifying as American Indian or Alaska Native	ACS

where x is the county-level value, and μ and σ are the statewide mean and standard deviation.

Domain scores were calculated as the unweighted mean of standardized indicators within each domain. A composite SDOH score was then computed as the mean of the six domain scores. Percentile ranks (0–100) were additionally calculated for each domain and for the composite score to support interpretable cross-county comparisons.

Table 3 summarizes the interpretation of domain-level measures. All domains are oriented such that higher values correspond to greater disadvantage. Positive z-scores indicate above-average levels of disadvantage relative to the state, while negative values indicate more favorable conditions. These measures are intended for descriptive and comparative purposes and do not represent absolute thresholds or causal relationships.

3.2.4. Dashboard Architecture and Analytic Modules

The computational system is implemented as an interactive R Shiny dashboard that serves as both an analytic interface and a guided learning environment for exploring SDOH through a Tribal lens. The dashboard integrates spatial data, SDOH indicators, and outcome measures within a unified platform designed to support both data exploration and user interpretation.

The system includes a dedicated landing page that provides an overview of the dashboard structure, key concepts, and guidance for navigation. The computational tool is developed, maintained, and owned by Cherokee Nation Public Health, and therefore prominently incorporates CNPH branding, including organizational identification within the title and interface. This design choice is intentional and aligns with principles of Indigenous Data Sovereignty, which emphasize local ownership, control, and governance of data systems. By situating the tool within the CNPH context, the system reinforces that the data, analyses, and interpretations are generated for and within the Tribal public health system, rather than as an externally developed analytic product. This introductory component is intended to orient users to the purpose of the tool, the structure of the SDOH domains, and the appropriate interpretation of indicators prior to engaging with the analytic modules.

Figure 3(A) illustrates the embedded recorded video tutorials that provide step-by-step instruc-

Table 3: Interpretation of SDOH Domains and Standardized Scores

Domain	Description	Interpretation of Higher Values
Economic Instability	Measures economic disadvantage, including poverty, unemployment, and reliance on public assistance	Higher values indicate greater economic disadvantage (worse conditions)
Education Attainment	Captures educational attainment levels, including proportions with low and high levels of education	Higher values indicate lower overall educational attainment (worse conditions)
Housing and Transportation Barriers	Reflects housing instability and transportation limitations, including cost burden, overcrowding, and lack of vehicle access	Higher values indicate greater housing and transportation barriers (worse conditions)
Health Care Access	Represents access to healthcare services, including insurance coverage and reliance on public or Tribal systems	Higher values indicate reduced access or greater barriers to care (worse conditions)
Environmental Burden	Captures environmental exposures and access to environmental resources	Higher values indicate greater environmental risk or reduced access to resources (worse conditions)
Social and Community Context	Reflects social structure and community characteristics, including disability, household composition, and demographic factors	Higher values indicate greater social vulnerability (worse conditions)
<i>Note: All indicators are standardized as z-scores prior to aggregation. Positive values indicate above-average levels of disadvantage relative to the state, while negative values indicate more favorable conditions. Domain scores and composite scores should be interpreted comparatively across counties.</i>		

tion on system use, interpretation of outputs, and key considerations related to data limitations and small-area analysis. These tutorials were developed using scripted content to ensure consistency and clarity across training materials. Full tutorial scripts are provided in Appendix A to support transparency and reproducibility of the training framework.

Figure 3(B) presents the dashboard landing page, which introduces the six SDOH domains that structure the analytic framework. These domains capture distinct but interrelated dimensions of social and environmental conditions influencing health, including economic, educational, housing, healthcare, environmental, and community factors. The landing page is designed to orient users to the domain-based structure of the data and to provide context for interpreting domain-level scores and comparisons across counties.

Beyond the introductory and training components, the dashboard includes multiple analytic modules that enable users to visualize, compare, and interpret SDOH patterns across counties, as shown in Figure 3(C). These modules support tasks such as spatial mapping (Maps), domain-level profiling (County Profile), clustering of counties with similar SDOH characteristics (Profiles/Clusters), exploratory assessment of relationships between SDOH domains and health outcomes (Tradeoffs), and an associative change framework (Scenario Sandbox). Together, these components form an integrated system that combines data access, analytic functionality, and user guidance within a single interface.

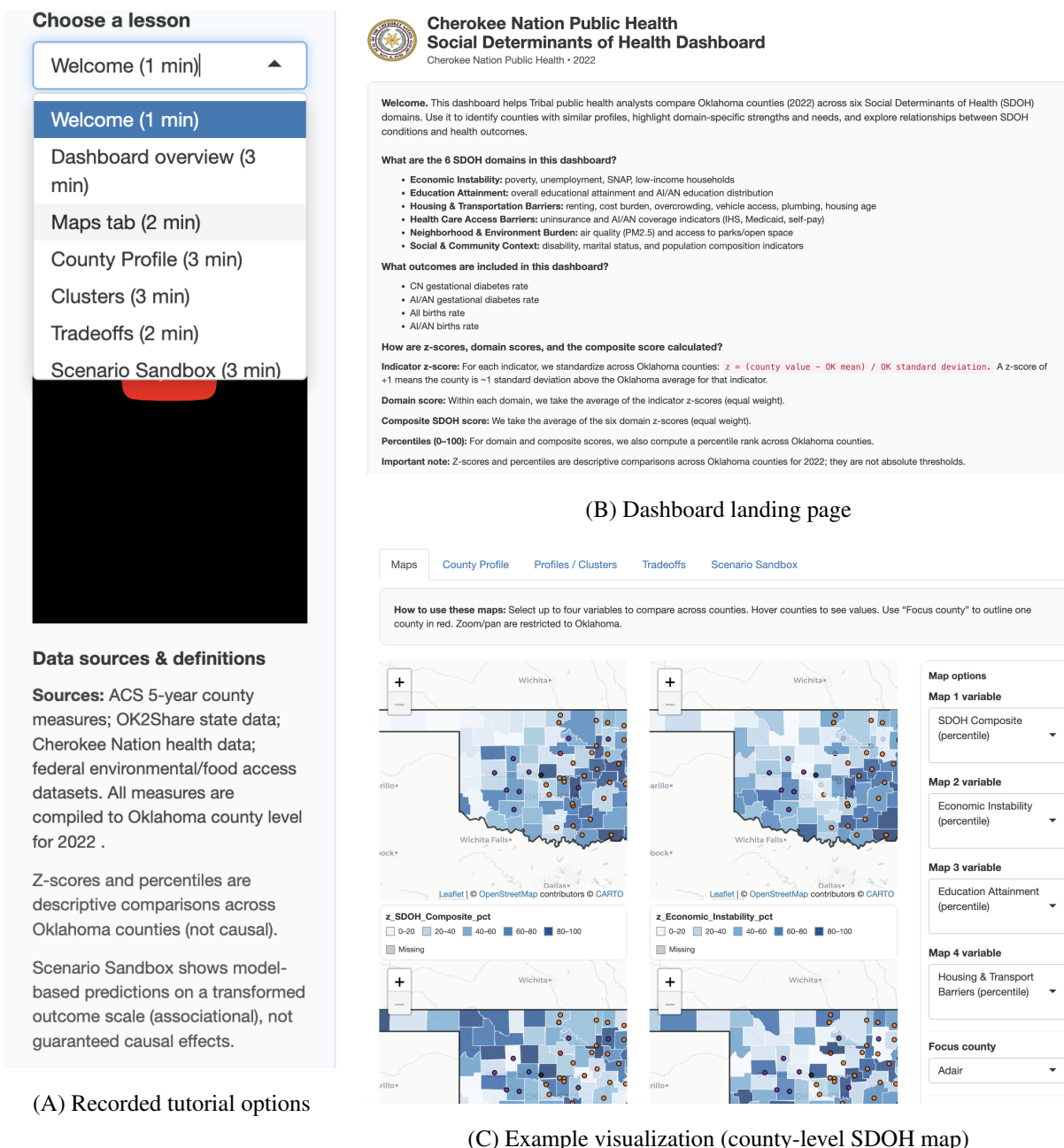


Figure 3: Overview of the Cherokee Nation SDOH dashboard interface. The sidebar (A) provides user controls for selecting video tutorials, while the main interface includes the landing page (B) and interactive visualization modules (C) for exploring county-level SDOH patterns.

Comparative Mapping: Figure 4 shows the multi-panel mapping interface, which allows users to visualize and compare multiple indicators or domain scores across counties simultaneously. Spatial visualization is a foundational approach in public health for identifying geographic patterns, disparities, and clustering of health-related conditions (Cromley and McLafferty, 2011;

Banerjee et al., 2014). In small-area contexts, such as county-level analyses, mapping provides an intuitive means of examining spatial heterogeneity that may not be apparent through tabular summaries alone (Wakefield, 2007; Rao and Molina, 2015).

The use of multiple synchronized map panels enables direct visual comparison across indicators or domains, supporting the identification of co-occurring patterns and potential spatial relationships. This comparative design aligns with established principles of visual analytics, which emphasize side-by-side displays as an effective strategy for facilitating pattern recognition and cognitive comparison across datasets (Schulze et al., 2023; Ware, 2012). By presenting multiple measures within a unified spatial frame, the interface supports exploratory assessment of geographic disparities and enhances interpretability for users engaged in applied public health decision-making.

Within the Tribal public health context, this functionality is particularly important for examining differences between counties with varying levels of AI/AN population representation and access to Tribal health systems. The mapping interface enables users to compare patterns across counties that include Tribal service areas, counties with higher proportions of AI/AN populations, and those with predominantly non-Tribal populations. This supports identification of spatial disparities that may reflect differences in access to care, socioeconomic conditions, and structural factors affecting Tribal communities. By facilitating these comparisons, the dashboard provides a tool for contextualizing SDOH patterns within the broader landscape of Tribal and non-Tribal populations, consistent with the goal of producing locally relevant and interpretable public health insights. An additional option within the mapping interface enables users to overlay point-level data representing Tribal health facility locations. Facilities are displayed using color-coded markers to distinguish between clinic types, including Indian Health Service and Tribal-operated facilities. This overlay provides a spatial reference for interpreting county-level SDOH indicators, allowing users to examine potential relationships between geographic access to Tribal health infrastructure and observed variation in social and health conditions. The integration of facility locations within the mapping interface enhances the interpretability of spatial patterns by situating SDOH measures within the context of healthcare access.

County Profile: Radar plots provide a multivariate summary of domain-level SDOH profiles for each county, allowing users to visualize relative strengths and disadvantages across domains and drivers within a single, interpretable display. Complementary driver tables quantify the contribution of individual indicators by identifying those that deviate most from statewide averages. Together, these components enable users to move from high-level domain patterns to specific underlying factors, supporting targeted interpretation of the conditions driving observed disparities. In the context of Tribal health, analysts can visualize how AI/AN-specific SDOH variables contribute to broader structural domains within a selected county.

Cluster Analysis: Counties were grouped using k-means clustering, an unsupervised partitioning algorithm that assigns observations to a pre-specified number of clusters by minimizing within-cluster variability (Hastie et al., 2009). In this context, each county is represented by a vector of domain percentile scores summarizing multidimensional SDOH conditions. The algorithm iteratively partitions counties into K clusters (K is selected by the user with options from 3 to 8) such that counties within the same cluster exhibit similar domain-level profiles, while those in different clusters are as distinct as possible. Cluster centroids represent the average SDOH profile within each group, facilitating interpretation of shared structural conditions across counties. A higher chosen K results in a more granular partitioning of counties, capturing finer distinctions in SDOH patterns, while a smaller K yields broader groupings that emphasize more general similarities.

In the context of Tribal public health, this clustering approach enables identification of counties that share similar configurations of structural conditions, rather than focusing on individual indicators in isolation. By examining patterns of co-occurring disadvantage—such as economic instability, limited healthcare access, and social vulnerability—the method provides a more holistic view of the environments shaping health outcomes in AI/AN populations. This is particularly important in settings characterized by small population sizes and fragmented data systems, where single indicators may be unstable or incomplete. Clustering supports comparison between counties with higher proportions of AI/AN populations or proximity to Tribal health systems and those with

similar or contrasting profiles, helping to identify groups of regions that may benefit from aligned public health strategies.

Tradeoff Analysis: Tradeoff analysis is implemented using bivariate comparisons of two selected measures, where counties are plotted within a two-dimensional space and classified into quadrants based on their relative standing (e.g., above or below the statewide median or mean). This quadrant-based approach enables identification of counties exhibiting contrasting patterns across indicators, such as high disadvantage in one domain but more favorable conditions in another. By jointly visualizing two dimensions, the method facilitates detection of potential tradeoffs, synergies, or discordant patterns that may not be apparent when indicators are examined independently (Heer et al., 2010; Ware, 2012).

In the context of Tribal SDOH surveillance, tradeoff analysis provides a structured way to examine how different dimensions of disadvantage co-occur or diverge across counties. This is particularly relevant in Tribal settings, where structural conditions such as economic instability, healthcare access, and environmental burden may not align uniformly. For example, counties with higher proportions of AI/AN populations may exhibit elevated disadvantage in certain domains while demonstrating relative strengths in others, reflecting the complex interplay between structural inequities and community assets. By identifying counties with contrasting profiles, this approach supports more nuanced interpretation of disparities and can inform targeted strategies that address specific combinations of needs rather than relying on single-indicator assessments.

Scenario Sandbox: The scenario sandbox is implemented as an interactive modeling component that allows users to explore how changes in SDOH domain scores are associated with changes in selected health outcomes. The underlying framework is based on linear regression models, where outcomes are modeled as a function of domain-level predictors. Users can simulate hypothetical changes in one or more domains and observe the corresponding predicted changes in outcomes, holding other factors constant. These predictions are intended for exploratory use and do not imply causal relationships, but rather illustrate associations observed in the data.

In Tribal public health contexts, the scenario sandbox provides a practical tool for exploring

potential implications of changes in structural conditions under data constraints. Given limitations in data completeness, measurement error, and the absence of fully specified causal models, this approach supports intuitive, scenario-based reasoning about how improvements or declines in specific SDOH domains may relate to health outcomes. This is particularly useful for engaging public health analysts in interpreting complex relationships and considering potential intervention priorities. By enabling users to interactively assess how different domains contribute to outcomes, the sandbox supports informed, contextually grounded decision-making while maintaining transparency about the non-causal nature of the underlying models.

3.3. Capacity-Building Framework

To evaluate whether the dashboard achieves its intended goals of usability, interpretability, and local capacity building, a mixed-methods assessment was conducted with CNPH analysts and is described in Section ???. The analytic system described above is complemented by an embedded capacity-building framework designed to support its meaningful and sustainable use. Rather than treating training as a supplementary activity delivered separately from the dashboard, the proposed framework integrates learning, interpretation, and use directly within the computational system. This reflects a deliberate design commitment to supporting locally governed, interpretable, and sustainable data practices within Tribal public health contexts.

In this framework, effective use of the dashboard requires not only access to data outputs, but also an understanding of how those outputs are generated, including underlying assumptions, data limitations, and analytic structure. This emphasis on transparency and interpretability is essential for supporting informed decision-making and aligns with principles of IDS and empowerment of local Tribal data stewards, which emphasize local authority, responsibility, and ethical use of data (Carroll et al., 2020; Rhodes et al., 2024). Our proposed integrated approach enables Tribal public health analysts to not only access and visualize data, but to critically interpret, adapt, and sustain analytic processes over time. By integrating analytic functionality with embedded training and guided interpretation, the system supports users in developing the conceptual and practical skills needed to engage with the “machine behind the tool,” rather than relying on opaque, externally

defined outputs.

3.3.1. Embedded Training Components

Training materials were embedded directly within the dashboard to support user learning in real time. These include short video tutorials that demonstrate key features of the system, including navigation, interpretation of SDOH indicators, and use of analytic modules.

In addition to technical instruction, training content addresses key conceptual topics, including data limitations, small-area instability, and differences across data sources. This approach reflects principles of adult learning theory, emphasizing problem-centered instruction and immediate applicability (Knowles et al., 2015; Kolb, 1984).

Table 4 provides a summary of the recorded self-guided training modules and the topics they cover.

3.3.2. User Feedback Session

To support iterative refinement and capacity building, user feedback sessions were conducted with Tribal public health analysts within Cherokee Nation Public Health. Participants engaged in structured, task-based interactions with the dashboard and provided feedback on usability, interpretability, and relevance to real-world workflows. Feedback was collected through guided discussion and structured prompts, as well as user-based surveys. Qualitative data were analyzed using inductive thematic analysis, in which themes were identified through iterative review of session discussion notes and survey responses, focusing on system functionality, clarity of visualizations, and practical utility.

These sessions were embedded within an iterative feedback loop in which user input directly informed subsequent modifications to both the dashboard and accompanying training materials. This cyclical process—consisting of user engagement, feedback collection, system refinement, and re-evaluation—is consistent with established principles of user-centered design and participatory development (Norman, 2013; International Organization for Standardization, 2019; Greenhalgh et al., 2017). Iterative refinement included updates to interface design, labeling, explanatory con-

Table 4: Integration of Analytic Modules and Embedded Training Components

Module	Analytic Function	Embedded Training and Interpretation Focus
Dashboard Overview	Introduces system structure and navigation	Discussion of data sources, types, and limitations; discussion of data fusion approaches between state/federal and Tribal data sources; orientation to SDOH domains; guidance on interpreting percentiles and z-scores
Comparative Mapping	Visualizes spatial patterns across counties and indicators	Interpretation of geographic variation, understanding spatial heterogeneity, and contextualizing AI/AN populations and Tribal health infrastructure
County Profile	Summarizes multidimensional SDOH conditions using domain scores and driver indicators	Interpretation of domain percentiles, identification of key drivers, and understanding how AI/AN-specific indicators contribute to broader domains
Cluster Analysis	Groups counties with similar SDOH profiles using k-means clustering	Interpretation of cluster patterns, understanding similarity across regions, and identifying structurally comparable counties
Tradeoff Analysis	Examines relationships between two indicators using quadrant-based classification	Identifying contrasting patterns across domains, interpreting co-occurring advantages and disadvantages, and recognizing multidimensional disparities
Scenario Sandbox	Explores associations between SDOH domains and health outcomes using linear models	Understanding non-causal interpretation, assessing domain-outcome relationships, and supporting scenario-based reasoning for decision-making

tent, and training modules to improve interpretability and usability.

In the context of Tribal public health systems, this feedback loop also serves as a mechanism for capacity building by actively engaging analysts in the development and interpretation of the analytic tool. Rather than positioning users as passive recipients of externally developed systems, this approach supports co-learning, enhances familiarity with data and methods, and promotes local ownership of the analytic workflow. Such participatory and iterative approaches align with principles of Indigenous Data Sovereignty, which emphasize local control, contextual interpretation, and sustained engagement in data processes (Rainie et al., 2019; Rhodes et al., 2024). By embedding training within an iterative development cycle, the workflow supports both immediate usability and long-term sustainability of data-driven decision-making in Tribal public health practice. Training materials developed for the Zoom-hosted user feedback sessions are provided in Appendix A. The user feedback survey instrument, administered via Google Forms, is provided in Appendix B.

3.4. Ethical Considerations

This study uses aggregated and de-identified data and was determined to be non-human subjects research by the Emory University Institutional Review Board (IRB). All activities were conducted in accordance with Tribal data governance principles, with particular attention to appropriate interpretation and use of data within the CNPH context.

3.5. Computational Tools

All data processing, analysis, and visualization were conducted using the R statistical computing environment (version 4.2.1) (R Core Team, 2023). R provides a flexible and extensible framework for data manipulation, statistical analysis, and reproducible workflows, and is widely used in public health and data science applications (Wickham, 2019; Wickham and Grolemund, 2017). The interactive dashboard was developed using the R Shiny framework, which enables the construction of web-based applications directly from R (Chang, 2021). This computational approach aligns with principles of reproducible research and transparent data workflows, enabling iterative development, scalability, and adaptation within Tribal public health systems (Wilkinson et al., 2016).

4. Results

4.1. Worked Example

To illustrate the use of the CNPH SDOH Dashboard, Figures 4–8 present a worked example for Adair County.

County Maps: Figure 4 displays the multi-panel comparative mapping interface, illustrating county-level SDOH patterns across the 77 Oklahoma counties. The illustrative example presents four domain percentile maps: (1) Education Attainment (top left), (2) Economic Instability (top right), (3) Housing & Transportation Barriers (bottom left), and (4) Health Care Access (bottom right). Each map includes point overlays representing Tribal health-affiliated clinics, with color indicating clinic type.

By comparing these spatial patterns, users can assess alignment and divergence across domains. For example, an inverse geographic relationship is observed between educational attainment and economic instability, particularly in southwestern Oklahoma, where counties with lower educational attainment tend to exhibit higher economic disadvantage. Focusing on the 14 Cherokee Nation counties in northeastern Oklahoma, these counties generally demonstrate moderate to high educational attainment alongside moderate to high economic instability, contrasting with broader statewide patterns where higher education is typically associated with lower economic disadvantage.

Additionally, comparison of clinic locations with the Health Care Access domain suggests that the distribution of Tribal health facilities does not directly correspond to measured access. This indicates that structural barriers to care—such as insurance coverage, transportation, or systemic constraints—may play a more substantial role than geographic proximity alone. Together, these spatial comparisons provide an entry point for identifying regional disparities and motivate further investigation using domain profiles and indicator-level drivers.

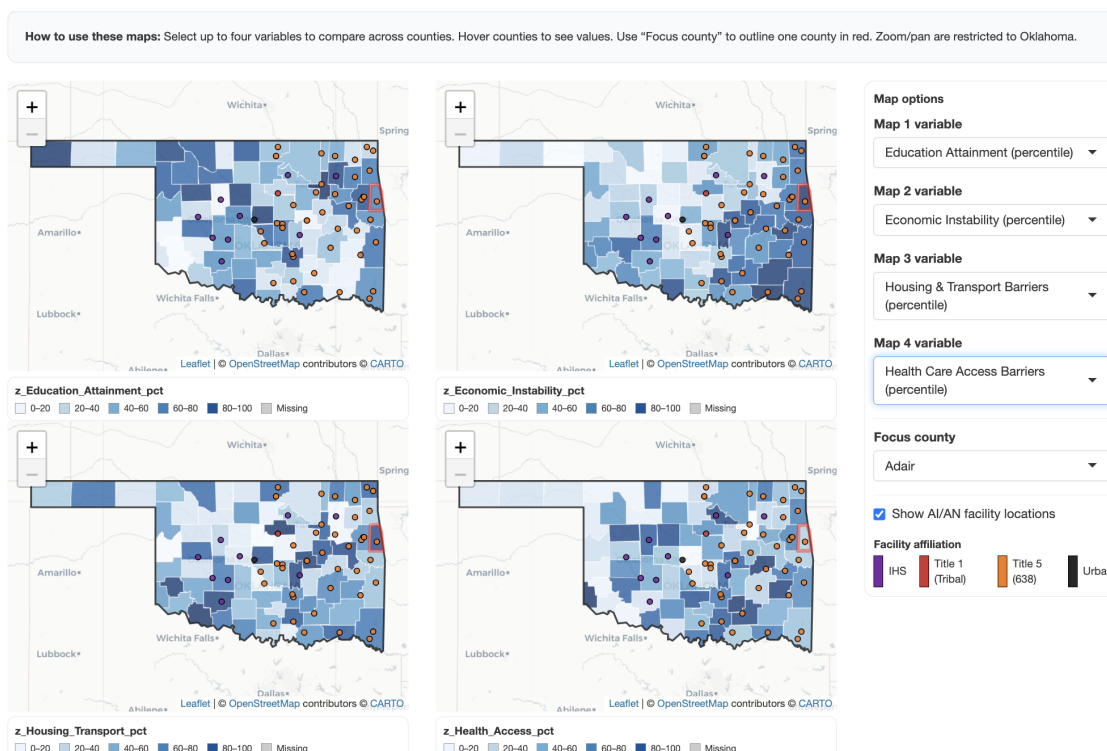


Figure 4: Example of multi-panel comparative mapping interface.

County Profile: Figure 5 illustrates the multidimensional SDOH profile of Adair County. The radar plot shows that Adair County has a higher-than-average AI/AN population aged 18–64 ($z = 4.4$), which is a major contributor to the elevated Social & Community Context domain percentile (93%). Additionally, Adair has a higher-than-average z-score for low-income households ($z = 2.5$), which is a primary driver of the elevated Economic Instability percentile (97%). Together, these patterns illustrate how county profiling enables users to move from broad domain-level summaries to the specific indicators contributing most strongly to observed disadvantage.

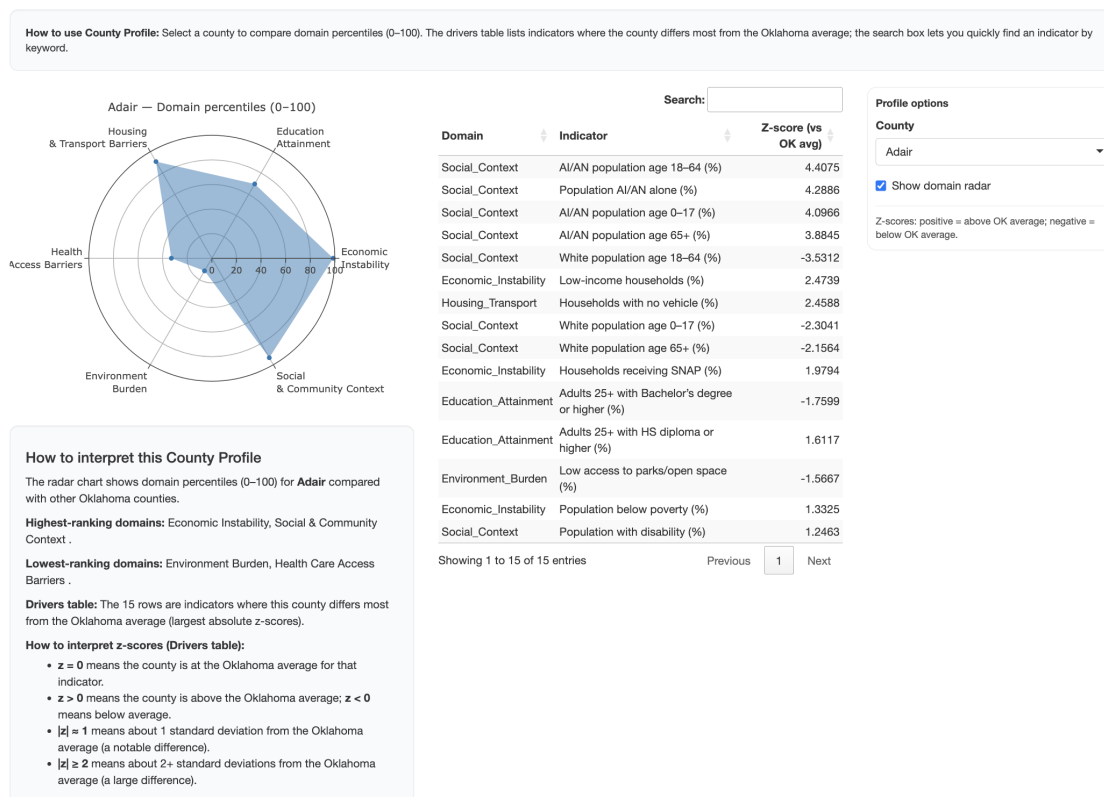


Figure 5: County profile visualization showing domain percentiles and key drivers.

Cluster Analysis: Figure 6 presents county groupings based on $k = 5$ clusters, revealing distinct regional patterns in multidimensional SDOH profiles. Spatially, several clusters exhibit geographic coherence, including concentrations in northwestern (blue), southeastern (yellow), and southwestern (green) Oklahoma, while others are more dispersed across central and northern regions.

Cluster profiles highlight meaningful differences in the configuration of structural conditions across counties. For example, Cluster 1 is characterized by high economic instability, low educational attainment, elevated housing and transportation barriers, and high social and community vulnerability, despite moderate levels of healthcare access and relatively lower environmental burden. This pattern reflects a combination of socioeconomic disadvantage and structural barriers that may jointly shape health outcomes. Adair County is classified within Cluster 5, which is characterized by comparatively lower economic instability and moderate to higher educational attainment, alongside elevated levels of social and community context vulnerability. This profile suggests that,

while certain economic and educational conditions may be more favorable relative to other clusters, underlying social and demographic factors remain important drivers of overall disadvantage.

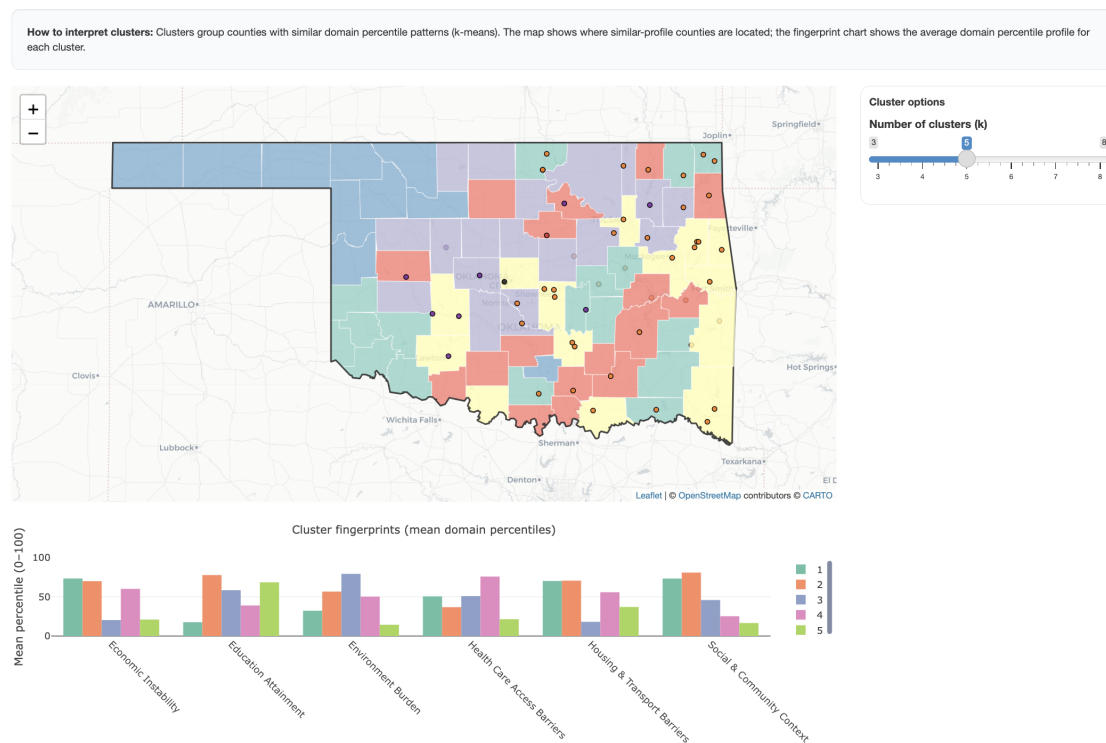


Figure 6: Cluster map and corresponding domain profile fingerprints.

Tradeoff Analysis: Figure 7 illustrates the bivariate relationship between Housing & Transportation Barriers (percentile) and the proportion of AI/AN population aged 18–64 across the 77 counties in Oklahoma. Counties are classified into four quadrants based on whether they fall above or below the statewide reference (e.g., median) for each measure: High–High, High–Low, Low–High, and Low–Low.

A clear geographic contrast emerges between eastern and western Oklahoma. Counties in the eastern region—where a larger proportion of Tribal populations reside—are predominantly characterized by elevated AI/AN population proportions, with variation in housing and transportation barriers ranging from moderate to high. In contrast, western counties are more frequently concentrated in the Low–Low and High Housing/Low AI/AN quadrants, reflecting lower AI/AN population representation and differing structural conditions.

This pattern highlights that counties with higher AI/AN population proportions are more likely to experience elevated or variable housing and transportation barriers, underscoring the importance of examining these dimensions jointly. More broadly, the tradeoff analysis demonstrates how structural conditions do not align uniformly across domains, and that disparities in Tribal regions may reflect distinct combinations of demographic composition and infrastructural constraints. This joint perspective supports more nuanced interpretation of SDOH patterns and helps identify counties with similar or contrasting profiles for targeted public health planning.

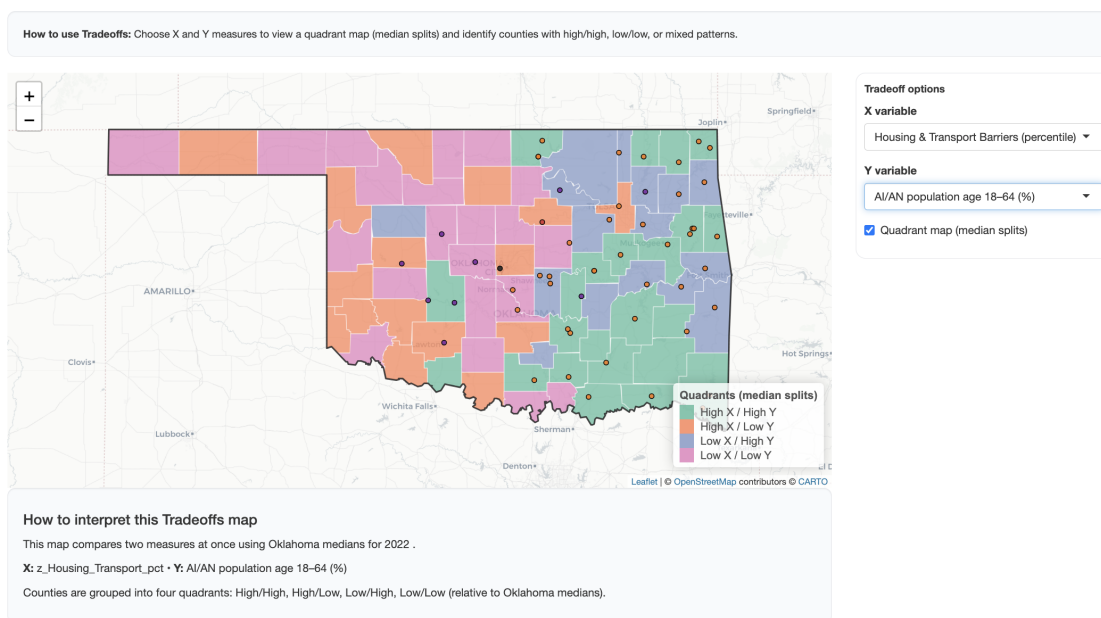


Figure 7: Tradeoff analysis using quadrant classification.

Scenario Sandbox: Figure 8 demonstrates the use of the Scenario Sandbox using Adair County as a worked example. In this scenario, educational attainment was increased from 0.08 to 1.38 (a change of +1.30 on the standardized scale). Under the fitted linear model, this increase is associated with an increase in the predicted Cherokee gestational diabetes rate on the log scale of approximately 5 units.

At the same time, the simulated change in educational attainment is associated with reductions in other domains, including economic instability (-0.37) and health care access barriers (-0.44). These results illustrate how changes in one domain are associated with shifts across multiple di-

mensions, reflecting the interrelated nature of SDOH conditions.

Importantly, these predictions are intended for exploratory interpretation and do not imply causal relationships. Rather, the sandbox provides a mechanism for examining how observed associations between domains and outcomes manifest under hypothetical changes, supporting intuitive understanding of complex, multidimensional relationships in Tribal public health contexts. The positive association between educational attainment and the modeled gestational diabetes outcome may reflect differences in diagnosis, reporting, or access to care rather than a direct causal relationship, further underscoring the need to interpret sandbox outputs within the broader context of data limitations and surveillance structure.

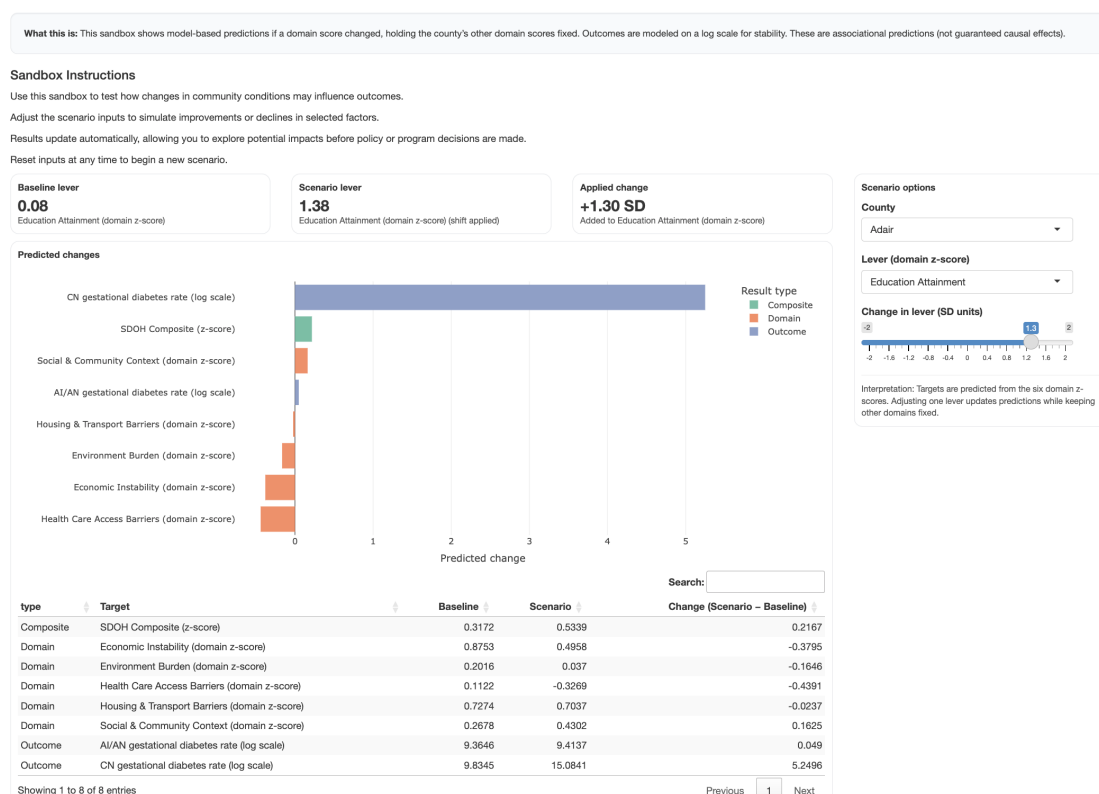


Figure 8: Scenario sandbox for exploratory analysis of domain-outcome relationships.

Taken together, these five analytical views provide a coherent and multi-layered picture of Adair County. The mapping module situates the county within the broader regional landscape, revealing that northeastern Oklahoma—where the majority of Cherokee Nation counties are located—exhibits a distinctive combination of moderate-to-high educational attainment alongside elevated

economic instability, a pattern that diverges from the statewide association between education and economic disadvantage. The county profile deepens this picture by identifying the specific structural drivers: an exceptionally high AI/AN population proportion and a high burden of low-income households. Cluster analysis confirms that Adair County belongs to a group characterized by elevated social vulnerability despite relatively more favorable economic conditions than the most disadvantaged clusters—suggesting that social and demographic factors are primary drivers of disadvantage, not economic conditions alone. Tradeoff analysis further shows that counties with the highest AI/AN population proportions tend to face compounding housing and transportation barriers, reinforcing the importance of intersecting structural conditions. Finally, the scenario sandbox—while strictly exploratory and non-causal—illustrates how changes in educational attainment propagate across multiple domains simultaneously, underscoring the interconnected nature of SDOH conditions in ways that single-indicator approaches cannot capture. This synthesis demonstrates the core value of the workflow: not any single module, but the progressive, interpretive chain from spatial pattern to structural driver to comparative typology to multidimensional tradeoff.

4.2. User Evaluation and Capacity-Building Insights

To evaluate the usability, interpretability, and practical relevance of the dashboard and associated training workflow, a mixed-methods assessment was conducted with Tribal public health analysts. Results are summarized across participant characteristics, quantitative usability ratings, and qualitative feedback from structured focus group sessions.

4.2.1. Participant Characteristics

The training and evaluation were conducted with five participants from Cherokee Nation Public Health (CNPH), including epidemiologists and data analysts. Participants reported between one and three years of experience in their roles and limited prior exposure to interactive dashboard-based analytic tools.

4.2.2. *Quantitative Findings*

Survey responses indicated overall positive usability of the dashboard. The mean overall system rating was 4.33 on a five-point Likert scale, reflecting a favorable user experience. Navigation and interaction features—including map exploration, zooming, and county selection—received the highest ratings (mean score 4.67), indicating that participants were able to effectively engage with the interface.

Understanding of SDOH domain concepts was also rated highly (mean score 4.50), suggesting that the domain-based framework was clearly communicated and interpretable. In contrast, interpretation of derived statistical measures, including z-scores, composite scores, and percentile rankings, received lower ratings (mean score 3.67), indicating moderate difficulty in understanding these concepts.

These results suggest that while the dashboard interface is accessible and intuitive, interpretation of standardized and derived metrics presents a greater challenge, particularly for users without formal statistical training.

4.2.3. *Qualitative Findings*

Qualitative feedback from user feedback sessions revealed four primary themes related to usability, interpretability, analytic functionality, and system scope.

Theme 1: Usability and Navigation. Participants consistently described the dashboard as intuitive and easy to navigate, even for users with limited prior experience with similar tools. The ability to explore multiple domains and interact with spatial visualizations was identified as a key strength supporting exploratory analysis.

Theme 2: Interpretation of Statistical Measures. Participants expressed uncertainty in interpreting standardized measures, particularly z-scores and composite domain scores. Questions arose regarding how z-scores were computed, how they contribute to composite domains, and how to interpret positive and negative values in a decision-making context. Feedback suggested the need for clearer explanation of statistical concepts, including simplified interpretation guides and categorical ranges (e.g., below average, above average).

Theme 3: Comparative and Analytic Functionality. Participants identified a need for enhanced comparative tools, including the ability to directly compare multiple counties within a single view (e.g., overlaying county profiles on radar charts). Additional suggestions included incorporating scatter plots alongside existing visualizations to support analysis of relationships between variables.

Theme 4: Data Scope and Contextual Relevance. Participants noted limitations related to geographic scope, particularly the restriction to Oklahoma counties, given that Cherokee Nation populations extend across state boundaries. Feedback also emphasized the importance of incorporating culturally relevant variables within domains such as Social and Community Context, as well as providing clearer documentation of data sources and underlying computations.

Additional feedback highlighted the importance of integrating training materials directly within the dashboard, including instructional videos and accessible explanations of data processing and analytic methods.

4.2.4. *Interpretation of Findings*

The evaluation results demonstrate that the dashboard and associated workflow are both feasible and usable within a Tribal public health setting. Participants were able to effectively navigate and interact with the system and identified clear value in its ability to support exploration of SDOH patterns across counties.

Importantly, the findings highlight a critical distinction between usability and interpretability. While interface interaction was rated highly, participants reported greater difficulty in interpreting standardized and derived statistical measures. This underscores that accessibility of analytic tools does not necessarily translate to interpretability of outputs, particularly in settings where users may have limited formal training in statistical methods.

These results emphasize the importance of integrating training directly within data systems, rather than treating analytic tools and user education as separate components. The observed challenges in interpreting derived metrics support the need for embedded explanatory content, simplified interpretation frameworks, and iterative learning mechanisms within the dashboard.

From a broader perspective, the findings reinforce the value of a workflow-based approach that combines data integration, visualization, and capacity building. The iterative feedback process—where user input informs system refinement—supports both immediate usability and longer-term sustainability. This approach aligns with principles of Indigenous Data Sovereignty, which emphasize local control, interpretability, and contextual relevance of data systems within Tribal public health practice. More broadly, these findings highlight a central challenge in public health data systems: improving access to data does not automatically improve interpretation. Addressing this gap requires integrating analytic workflows with embedded training and user-centered design, particularly in Tribal public health contexts.

5. Discussion

This thesis developed and evaluated an applied data science workflow for strengthening Tribal public health surveillance through the integration of multi-source data, interactive visualization, and embedded training within a single analytic system. In collaboration with Cherokee Nation Public Health (CNPH), the project produced a county-level Social Determinants of Health (SDOH) dashboard designed to support exploratory analysis, contextual interpretation, and locally relevant decision-making. Taken together, the findings demonstrate that effective public health data systems in Tribal contexts require more than data integration and visualization alone. Rather, they must also support interpretability, local capacity building, and iterative refinement in order to be sustainable and useful in practice.

A central contribution of this work is the development of a workflow-based approach that addresses a persistent gap in the literature between methodological innovation and real-world implementation. Existing statistical and epidemiologic literature has identified substantial challenges in producing reliable estimates for Tribal populations, including small population sizes, suppression of publicly released estimates, racial misclassification, and fragmentation across Tribal, state, and federal data systems. While advanced analytic methods exist for addressing sparsity, bias, and multi-source integration, these approaches are often difficult to implement within routine public health practice due to limitations in data access, computational infrastructure, and local analytic capacity. The present work contributes to this gap by demonstrating an alternative pathway: a transparent, descriptive, and operationally feasible system that supports data use under real-world constraints while remaining aligned with principles of Indigenous Data Sovereignty (IDS).

The worked example for Adair County illustrates the value of this framework for multidimensional interpretation of SDOH patterns within a Tribal public health context. Across the dashboard modules, the system supported several complementary levels of interpretation. The mapping interface situated Adair County within broader statewide and regional spatial patterns, allowing users to assess whether observed conditions reflected localized disadvantage or broader regional structure. The county profile module then identified specific drivers underlying elevated domain scores,

showing how AI/AN population structure and low-income household burden contributed to the county's social and economic vulnerability. Cluster analysis placed Adair County within a broader typology of structurally similar counties, while tradeoff analysis highlighted how demographic composition and infrastructural disadvantage co-occurred across eastern Oklahoma. Finally, the scenario sandbox provided an exploratory mechanism for examining how changes in one domain may be associated with shifts in other domains and outcomes. Together, these components demonstrate that the dashboard functions not merely as a visualization platform, but as an interpretive workflow for understanding complex and interconnected structural conditions.

At the same time, the evaluation findings make clear that usability and interpretability are distinct dimensions of effective data systems. Participants reported that the dashboard was intuitive and easy to navigate, and quantitative ratings indicated strong performance for interaction features such as map exploration and county selection. However, participants also reported greater difficulty interpreting derived statistical measures, particularly z-scores, composite indices, and percentile rankings. This distinction is one of the most important findings of the study. Public health dashboards are often evaluated primarily in terms of usability, accessibility, or technical functionality. However, the present findings suggest that an interface may be easy to use while still failing to support confident interpretation of analytic outputs. In other words, improving access to data does not automatically improve analytic understanding.

This finding has important implications for the design of Tribal public health data systems. In contexts where users may not have formal statistical training, the interpretability of standardized and derived metrics cannot be assumed. Instead, interpretability must be intentionally designed into the system through explanatory content, guided examples, training materials, and iterative opportunities for user engagement. In this project, the integration of recorded tutorials, structured focus group sessions, and user feedback mechanisms represents a move toward this broader conception of analytic usability. The results suggest that training should not be treated as an add-on to dashboard deployment, but as a core component of the system itself. This is particularly relevant in Tribal public health settings, where long-term sustainability depends not only on tool availability,

but on whether local analysts can confidently interpret, adapt, and maintain the analytic process over time.

A second important contribution of this work is its emphasis on local governance and context. The dashboard was intentionally developed within the CNPH setting, with Cherokee Nation branding, Tribal-relevant indicators, and iterative input from Tribal public health analysts. This design is substantively important. Much of the existing analytic infrastructure relevant to Indigenous populations is developed outside of Tribal systems, often by state agencies, federal institutions, or academic researchers. Although such systems may produce useful data, they do not necessarily align with Tribal priorities, governance structures, or decision-making contexts. By contrast, this project positions the dashboard as a locally situated analytic resource that supports Tribal interpretation of both publicly available and Tribal-contextualized data. In this way, the work extends IDS from a conceptual framework to a practical design principle for public health analytics.

The findings also point to the importance of comparative and contextual analytic functionality for Tribal public health users. Participants expressed interest in being able to compare counties directly, overlay county profiles, and examine relationships between continuous variables using additional plots such as scatterplots. These requests are informative because they indicate that users do not simply want to view static summary measures; they want to conduct comparative reasoning across places, domains, and structural conditions. This reflects how public health analysis is often conducted in practice, particularly in settings where analysts are trying to understand whether a pattern is unique to one county, shared across a region, or reflective of broader systemic inequities. The dashboard modules developed here begin to support that form of reasoning, but the feedback suggests that future versions could strengthen this comparative functionality further.

The evaluation additionally highlighted the importance of geographic scope and representational adequacy. Participants noted that restricting the dashboard to Oklahoma counties limits its utility in the context of Cherokee Nation populations that extend across state boundaries. This observation reinforces a larger issue in Indigenous public health data systems: administrative boundaries used in publicly available data often do not map cleanly onto Tribal geographies, service

areas, or lived community relationships. Although county-level aggregation provided a practical and interpretable unit for the present study, future work should explore ways to incorporate broader service regions, cross-state comparisons, or more locally meaningful spatial representations where feasible. Doing so would further align the analytic framework with Tribal governance and population realities.

Several limitations should be acknowledged. First, the dashboard is based primarily on county-level aggregation, which necessarily obscures within-county heterogeneity and may mask important local variation. This was an intentional design choice motivated by data availability, stability, and interpretability, but it limits fine-scale inference. Second, the system is descriptive and exploratory rather than inferential. The dashboard was not designed to provide causal estimates or formally correct for all sources of bias in the underlying data. In particular, the scenario sandbox is based on linear regression models and is intended only to illustrate associations under hypothetical changes. As noted in the worked example, some associations may reflect surveillance structure, diagnosis patterns, or reporting differences rather than substantive causal effects. Third, the evaluation sample was small and drawn from a single Tribal public health setting. While this is appropriate for an implementation-oriented formative evaluation, it limits generalizability beyond the immediate Cherokee Nation context. Fourth, because the dashboard integrates multiple data sources with differing coverage and definitions, some constructs remain imperfect proxies for the underlying social and structural processes of interest.

These limitations, however, do not diminish the value of the proposed framework. Rather, they reflect the realities of data-constrained public health settings and underscore the need for approaches that are transparent about those constraints. One of the strengths of this work is that it does not present the dashboard as a complete or final solution to Tribal data inequities. Instead, it offers a practical model for how analytic systems can be designed to operate responsibly within fragmented environments while remaining useful to local analysts. In that sense, the workflow is both a product and a process: a product in the form of the dashboard itself, and a process in the form of iterative refinement, user engagement, and training-centered implementation.

There are several important directions for future work. From a technical perspective, future versions of the dashboard could incorporate additional comparative features, improved explanatory aids for standardized metrics, more flexible visualization options, and expanded geographic coverage. More advanced modeling components could also be integrated where appropriate, particularly if additional data linkage and computational infrastructure become available. From an implementation perspective, future work should evaluate how the dashboard is used over time in routine public health workflows, whether embedded training improves confidence and interpretation, and how such systems can be maintained within Tribal institutions beyond the initial development period. Comparative work across multiple Tribal public health settings would also be valuable for understanding which components of the workflow are broadly transferable and which require adaptation to local governance, data access, and population context.

In conclusion, this thesis demonstrates that strengthening Tribal public health surveillance requires a broader conception of analytic infrastructure—one that includes not only data integration and visualization, but also training, interpretability, local ownership, and iterative design. By developing and evaluating a dashboard-centered workflow in collaboration with Cherokee Nation Public Health, this work provides a practical and theoretically grounded model for supporting data-driven decision-making under conditions of fragmentation and limited analytic infrastructure. More broadly, it suggests that the future of equitable public health data systems in Tribal contexts lies not in externally imposed analytic products, but in locally governed workflows that make complex data interpretable, usable, and sustainable in practice.

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Appendix A. User Training Protocol

Appendix A.1. Overview

This appendix presents the full training protocol used for the focus group session with Cherokee Nation Public Health analysts. The purpose of this document was to provide a standardized guide for introducing participants to the Social Determinants of Health (SDOH) dashboard, walking them through its major features, and facilitating structured discussion during the training session. The protocol included an introduction to the session, explanations of the dashboard structure and analytic concepts, guided demonstrations of each major dashboard component, interactive exercises, and a transition to the user feedback survey.

Appendix A.2. Training Guide

Section I. Introduction

1.1 Welcome. Hello everyone. On behalf of my professor, Emily Peterson, it is a pleasure to welcome you to today's training on the Cherokee Nation Public Health Dashboard.

1.2 Presenter Introduction. Before we begin, I would like to briefly introduce myself and explain how I came to this program.

My name is Joseph Ali, and I am a final-year student at the Emory University Rollins School of Public Health. I am in the Hubert Department of Global Health, where I am pursuing an MPH in Global Health along with the Data Science Certificate.

I became part of this program through my personal interest in working with populations facing disproportionate health outcomes, as well as my interest in data science and the development of data visualization tools that can help address health disparities. Through that path, I was introduced to Professor Emily Peterson, and that brings me here today.

1.3 Participant Introductions. I know many of you already know each other, but for my sake, so that I can relate better with everyone, I would appreciate it if each of you could please share your name and your role within Cherokee Nation Public Health.

1.4 Recording Permission. Before we continue, I would also like to ask for your permission to record this session. The purpose of the recording is to help us capture your insights so we can improve the dashboard. This recording will not be shared outside of our team, and your responses will remain confidential. Do I have your permission to record? Thank you.

1.5 Session Outline. Let me now walk us through the outline for today's training.

In this session, there will be two parts:

1. A live walkthrough of the dashboard, followed by interactive discussion and questions.
2. A short user feedback survey.

Section II. Dashboard Overview

2.1 Purpose of the Dashboard. This platform was designed to help Tribal public health scientists explore and better understand the social and environmental conditions that shape health in Tribal communities. Health is not determined by health care alone; it is also influenced by where people live, learn, work, and interact with their environment.

2.2 How the Dashboard Should Be Used. It is important to note that this dashboard reflects relative patterns within the dataset and is intended for analytic and planning purposes. While the data are publicly sourced and methodologically standardized, they should always be interpreted within a broader local context.

This dashboard is also meant to provide you with additional independence and resources to design and conduct analyses, support your work, and help validate how federal data reflect Tribal populations in Oklahoma.

2.3 Relevance for Tribal Public Health Analysts. This leads us into the broader discussion of Social Determinants of Health, why they matter, and how this dashboard is relevant to you as Tribal public health analysts.

We will upload a link in the chat for everyone to access the dashboard. Make sure everyone can open it so that they follow along.

2.4 Dashboard Layout. On the main page of the dashboard, on the left-hand side, there is a tool that allows you to watch training videos on how to use the different components of the dashboard. You simply click on the dropdown menu and select the video you want to watch.

There is a separate video for each major feature, and each one walks you through both how to use the tool and how to interpret the results.

To the right of that, there is a navigation bar showing the major dashboard tools: Maps, County Profile, Profiles/Clusters, Tradeoffs, and Scenario Sandbox. I will demonstrate each of these tools as we move through the training.

Section III. SDOH Definitions and Outcomes

3.1 Six SDOH Domains. To capture the multidimensional nature of Social Determinants of Health, we combined individual SDOH indicators into six broader domains:

1. Economic Instability
2. Education Attainment
3. Housing and Transportation Barriers
4. Health Care Access Barriers
5. Environmental Burden
6. Social and Community Context

3.2 How the Domain Scores Are Calculated. Each individual variable, such as the county-level unemployment rate, is first standardized into a z-score so that all variables are placed on a common scale and can be compared across indicators.

These z-scores are then combined to calculate domain scores, which are defined as the average of the z-scores across the variables within a domain.

A composite SDOH score is calculated as the average across the six domains.

Percentile rankings are also computed across the 77 counties in Oklahoma.

3.3 Health Outcomes Included in the Dashboard. The health outcomes included in the dashboard are:

1. Cherokee Nation gestational diabetes rate reported from Cherokee Nation health clinics,
2. AI/AN gestational diabetes rate reported by OK2Share,
3. all births rate, and
4. AI/AN births rate.

3.4 Transition to the Dashboard Walkthrough. Now that we have reviewed the six SDOH domains and the outcomes included in the dashboard, let's begin exploring how these patterns vary across all 77 counties in Oklahoma.

Appendix B. User Feedback Survey Instrument

The following survey was administered to Cherokee Nation Public Health (CNPH) participants via Google Forms immediately following the dashboard training session. The instrument was designed to capture structured feedback across four domains: participant background, dashboard usability, data clarity, and perceived usefulness for tribal public health practice.

Cherokee Nation Public Health Dashboard — User Feedback Survey

Purpose: This survey collects feedback on the usability, clarity, and usefulness of the Cherokee Nation Public Health Social Determinants of Health (SDOH) Dashboard. Your responses will help improve the dashboard so it better supports Tribal Public Health analysts and decision-making. Please answer based on your experience using the dashboard during this session.

Section 1: Background

Q1. Before today's session, had you previously seen or used this dashboard?.

- Yes, I have used it before
- This is my first time seeing it

Q2. What is your role at Cherokee Nation Public Health?.

- Tribal Public Health Analyst
- Epidemiologist
- Medical Director
- Program Manager
- Data Analyst
- Student/Intern
- Other: _____

Q3. How long have you worked in your current role?.

- Less than 1 year
- 1–3 years
- 3–5 years
- 5–10 years
- More than 10 years

Q4. How familiar are you with Social Determinants of Health (SDOH)?.

- Very familiar
- Somewhat familiar
- Slightly familiar
- Not familiar

Q5. How familiar are you with navigating public health dashboards?.

- Very familiar
- Somewhat familiar
- Slightly familiar
- Not familiar

If familiar, which dashboards have you used? (optional): _____

Section 2: Dashboard Usability

Q6. How easy was it to access the dashboard?.

- Very easy Easy Neutral Difficult Very difficult

Q7. How easy was it to navigate the dashboard overall?.

- Very easy Easy Neutral Difficult Very difficult

Q8. How clear were the instructions and explanations provided in the dashboard?.

- Very clear Clear Neutral Unclear Very unclear

Feature	Very Easy	Easy	Neutral	Difficult	Very Difficult
Viewing maps	☐	☐	☐	☐	☐
Comparing maps	☐	☐	☐	☐	☐
Zooming and panning	☐	☐	☐	☐	☐
Identifying counties	☐	☐	☐	☐	☐
Using the County Profile tab	☐	☐	☐	☐	☐
Understanding clusters	☐	☐	☐	☐	☐
Using the Tradeoffs tab	☐	☐	☐	☐	☐
Using the Scenario Sandbox	☐	☐	☐	☐	☐

Q9. How easy was it to perform the following tasks in the dashboard?.

Section 3: Understanding the Data

Q10. How clear were the six SDOH domains presented in the dashboard?.

- Very clear Clear Neutral Unclear Very unclear

Q11. Which domains were easiest to understand? (Check all that apply).

- ☐ Economic Stability
- ☐ Education Attainment
- ☐ Housing & Transportation
- ☐ Health Care Access
- ☐ Environmental Burden
- ☐ Social & Community Context

Optional: Why? _____

Q12. Which domains were hardest to understand? (Check all that apply).

- ☐ Economic Stability
- ☐ Education Attainment
- ☐ Housing & Transportation
- ☐ Health Care Access
- ☐ Environmental Burden
- ☐ Social & Community Context

Optional: Why? _____

Concept	Very Clear	Clear	Neutral	Unclear	Very Unclear
Z-scores	☐	☐	☐	☐	☐
Domain scores	☐	☐	☐	☐	☐
Composite SDOH score	☐	☐	☐	☐	☐
Percentiles	☐	☐	☐	☐	☐

Q13. How clear were the following concepts used in the dashboard?.

Q14. How easy was it to compare counties using the dashboard?.

- Very easy Easy Neutral Difficult Very difficult

Section 4: Usefulness for Public Health Work

Q15. How useful do you think this dashboard could be for your work?.

- Very useful Useful Neutral Not very useful Not useful

Q16. What are the most useful mechanisms in the dashboard?. _____

Q17. What improvements would you suggest for the dashboard?. _____

Section 5: Overall Feedback

Q18. Overall, how would you rate the dashboard?.

- 1 (Poor) 2 3 4 5 (Excellent)

Q19. What additional features or tools would you like to see added?. _____

Q20. What types of Tribal public health questions would you want this dashboard to help answer?.

Appendix C. User Feedback Analysis Report

Overview

This report summarizes feedback collected from Cherokee Nation Public Health (CNPH) staff following a structured dashboard training and usability session conducted on March 9, 2026. Three CNPH epidemiologists completed the post-training survey (all first-time users of the dashboard), and all four session participants contributed to the focus group discussion. Participants had 1–3 years of experience in their current roles. The evaluation examined dashboard usability, conceptual clarity of SDOH data displays, and perceived usefulness for tribal public health practice.

Quantitative Survey Results

Responses to Likert-scale items were coded numerically (1 = Very Difficult/Very Unclear/Not Useful; 5 = Very Easy/Very Clear/Very Useful) to enable calculation of mean scores across respondents.

Overall Ratings

All three respondents rated the dashboard favorably. The mean overall rating was 4.33 out of 5.0, with two respondents assigning a score of 4 (“Good”) and one assigning a score of 5 (“Excellent”).

Dashboard Usability

Table C.5 presents mean scores for usability items. Dashboard access and navigation were rated highly (means of 4.33 and 4.67, respectively), reflecting that participants found the interface approachable despite having no prior exposure. Clarity of in-dashboard instructions received a lower mean score of 3.67, indicating that explanatory text within the interface warrants revision.

Map-related tasks (viewing, zooming/panning) received the highest task-level scores (mean = 4.67), consistent with participants reporting that county comparison and geographic identification were among the most intuitive features. More advanced analytical tools—the Tradeoffs tab and Scenario Sandbox—received lower scores (mean = 4.00), suggesting that these features benefit from additional in-session explanation or guided examples.

Table C.5: Mean usability ratings by survey item (scale: 1 = Very Difficult to 5 = Very Easy or Very Clear; $n = 3$)

Survey Item	Mean Score
<i>Overall Access and Navigation</i>	
Ease of accessing the dashboard	4.33
Ease of navigating the dashboard overall	4.67
Clarity of in-dashboard instructions	3.67
<i>Tab-Specific Tasks</i>	
Viewing maps	4.67
Comparing maps	4.33
Zooming and panning	4.67
Identifying counties	4.00
Using the County Profile tab	4.33
Understanding clusters	4.33
Using the Tradeoffs tab	4.00
Using the Scenario Sandbox	4.00

Conceptual Clarity of SDOH Metrics

Table C.6 presents mean clarity ratings for the four statistical concepts displayed throughout the dashboard. Z-scores achieved perfect clarity ratings across all three respondents (mean = 5.00), likely reflecting the focus group discussion in which z-score computation and interpretation were addressed explicitly before survey completion. Domain scores and composite SDOH scores received means of 4.33, while percentiles averaged 4.67.

Table C.6: Mean conceptual clarity ratings by metric (scale: 1 = Very Unclear to 5 = Very Clear; $n = 3$)

Concept	Mean Score
Z-scores	5.00
Percentiles	4.67
Domain scores	4.33
Composite SDOH score	4.33

Perceived Usefulness

All three respondents selected “Very useful” when asked how useful the dashboard could be for their work (mean = 5.00). County comparison and geographic identification were cited as the most immediately applicable features for routine epidemiological tasks.

Qualitative Findings from the Focus Group Discussion

The focus session generated in-depth discussion organized around five technical topics: z-score and percentile computation, the Social and Community Context domain, the County Profile tab, cluster analysis, and data sources and training needs. Four themes emerged from inductive analysis of the session transcript.

Theme 1: Tension Between Statistical Precision and Decision-Maker Accessibility

Participants engaged substantively with the statistical architecture of the dashboard, asking how z-scores were centered, whether the reference population was the full 77-county Oklahoma sample or only the 14 AI/AN-majority counties, and how cancellation effects within composite domain scores could obscure meaningful variation. The group's proposed resolution—replacing numeric z-scores with a labeled categorical scale (e.g., ≥ 1 = “Much better than average”; -1 to 0 = “Below average”)—reflects a broader concern that statistical summaries must be legible to program managers and community representatives, not only analysts. A complementary recommendation was to maintain full z-score tables in a downloadable technical appendix within the dashboard, preserving analytical depth for advanced users without cluttering the primary interface.

Theme 2: Comparative Epidemiology as a Core Use Case

Across multiple discussion items, participants returned to the need for side-by-side county comparison. The County Profile tab's radar chart currently displays one county at a time; participants requested an overlay feature allowing two county profiles to be rendered simultaneously. One respondent's written survey comment reinforced this theme: “*A way to compare two counties on the county profiles tab.*” This finding aligns with the broader epidemiological workflow at CNPH, where identifying “peer counties” with similar SDOH profiles is central to program targeting and resource allocation decisions.

Theme 3: Geographic Boundaries Do Not Reflect Tribal Jurisdiction

One participant raised a substantive concern about the dashboard's geographic scope: the Cherokee Nation's service population is not bounded by the Oklahoma state border, and contigu-

ous counties in Arkansas (particularly Washington County, AR) and Kansas share more SDOH characteristics with CNPH-served populations than some western Oklahoma counties included in the current analysis (e.g., Cimarron County). A second participant extended this concern to other tribal nations, suggesting that the dashboard should eventually accommodate SDOH data for tribal populations across state lines. The group's recommendation was to expand the dashboard's geographic frame to include Oklahoma, Kansas, and Arkansas counties, an approach that would also partially address the small-numbers problem affecting the Scenario Sandbox's predictive module (currently limited to seven Cherokee Nation counties for gestational diabetes outcomes).

Theme 4: Training Infrastructure as a Sustainability Requirement

Participants noted that all session attendees were first-time users, yet the dashboard's core navigation features were rated 4.33–4.67 out of 5.00. This suggests that the training session design was effective for basic usability. However, the discussion identified gaps that a single training session cannot fully address: the statistical rationale for composite domain construction, culturally appropriate variable selection for the Social and Community Context domain, and the limitations of standard regression in the Scenario Sandbox for small tribal geographies. Participants recommended embedding short training videos directly within the dashboard interface so that new users can access methodological explanations asynchronously, supporting institutional capacity without requiring repeated in-person sessions.

Summary of Recommendations

Based on the combined quantitative and qualitative evaluation, the following improvements are recommended for subsequent dashboard iterations:

1. **Z-score display:** Add a categorical interpretation scale alongside z-score values throughout the dashboard; retain technical z-score tables in a downloadable supplement.
2. **County Profile tab:** Implement a two-county radar chart overlay to support direct domain-level comparison.

3. **Geographic scope:** Extend the dashboard to include contiguous counties in Arkansas and Kansas that fall within the Cherokee Nation's social and jurisdictional sphere.
4. **In-dashboard instructions:** Revise explanatory text accompanying composite domain scores and percentiles to improve accessibility for non-technical users (target: clarity score ≥ 4.50).
5. **Embedded training materials:** Develop and host short instructional videos within the dashboard, covering z-score interpretation, cluster analysis, and Scenario Sandbox use.
6. **Scatter plot integration:** Add scatter plots to the cluster analysis tab to visualize continuous relationships between SDOH variables and support exploratory analysis.
7. **Cultural scope of Social and Community Context:** Convene a CNPH working group to review and refine the variables constituting this domain to ensure they capture structurally relevant indicators of Indigenous community context.